

Contextuality-by-Default Analysis of S&P 500 Crisis Periods

Design, Mathematics, and the Post-Selection Interpretation
CRSP/WRDS daily data

Bell / stocks project — analysis specification

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1 Objective and framing

This document specifies a complete Contextuality-by-Default (CbD) analysis of S&P 500 constituent pairs over crisis and calm periods, using CRSP daily data from WRDS. The analysis has three deliverables:

1. a working CbD pipeline layered on the existing four-*S* (CHSH) machinery;
2. a *deflation result*: a demonstration that apparent Bell-type violations in equity data are reproducible by classical processes once the endogenous definition of the measurement contexts is accounted for; and
3. a *behavioral characterization*: how the dependence structure differs between crisis and calm regimes — the aggregate System-1/System-2 contrast — stated as classical regime-dependence, not contextuality.

★ What we are testing, and what we are not

We are *not* hunting for “quantum stocks.” The structural problem (§5) is that in market data the measurement *conditions* are computed from the same numbers as the measurement *outcomes* — like labeling years “drought years” using the harvest itself and then claiming drought explains the harvest. Because of this, a positive contextuality score here cannot certify anything non-classical. The goals are therefore to show this explicitly, quantify it, and extract the real (classical) crisis/calm structure — while the genuine contextuality question is reserved for decision experiments where conditions are set externally.

2 Data: CRSP via WRDS (build plan)

This is a verified, build-ready specification. Two practical facts dominate it. First, the historical S&P 500 constituent list is *no longer available through Compustat*: S&P withdrew the old `comp.idxcst_his` table from WRDS, so index membership must now come from CRSP. Second, CRSP migrated to its Flat File Format 2.0 (“CIZ”) in 2022; the legacy (SIZ/FIZ) tables were frozen after the December 2024 data cut, and the CIZ `_v2` tables are the sole forward source. All table names below are the current CIZ ones.

△ Two routes that will silently fail

- (1) `comp.idxcst_his` — the old Compustat index-constituents table; S&P pulled it from the WRDS platform, so it no longer returns S&P 500 membership. Use the CRSP list instead.
- (2) Legacy CRSP tables *without* the `_v2` suffix are frozen at the December 2024 cut and receive no new data; any sample reaching past 2024 must use the `_v2` (CIZ) tables. Both failure modes are quiet — empty or stale results, not errors.

Access. Use the `wrds` Python package over the WRDS PostgreSQL connection; every query below is a `db.raw_sql(...)` call. Because CIZ renamed many columns, confirm exact field names once with `db.describe_table(...)` before relying on them.

```
import wrds
db = wrds.Connection(wrds_username="...")    # one-time .pgpass setup
db.list_libraries()                        # confirm access
db.describe_table("crsp", "dsf_v2")        # confirm CIZ column names
```

2.1 Point-in-time S&P 500 membership

Historical members of the index (SPX) are in `crsp.dsp500list_v2` (daily; the schema-qualified form is `crsp_a_indexes.dsp500list_v2`). Each row is one membership spell: `permno`, `mbrstartdt` (member start date), `mbrenddt` (member end date). A security is a constituent on date D iff $mbrstartdt \leq D \leq mbrenddt$.

```
spx = db.raw_sql("""
    select permno, mbrstartdt, mbrenddt
    from crsp.dsp500list_v2
    """)
```

Applying this spell logic to *each* window so the universe is the index composition *as of that date* is what removes survivorship and look-ahead bias — using today’s index to study 2008 would silently condition on survival.

△ Membership-date conventions (subtle, costly to miss)

The end date of a *currently active* member defaults to the last trading day of the latest data cut and is revised forward at each annual update — so current membership is right-censored, not permanent. Because the list is daily, a membership test written against a *non-trading* day (e.g. Dec 31) can return nothing; anchor tests to trading days. Always join on `permno`, never on ticker: tickers are reused and reassigned (Facebook→Meta is one of many).

2.2 Daily returns and identifiers

Daily returns are in `crsp.dsf_v2` (the CIZ daily stock file), keyed by `permno` and `date`. The holding-period return `ret` is adjusted for splits and distributions, and — a change from the legacy workflow — in CIZ the *delisting return is already folded into the main return series*, so no separate delisting-file merge is required.

```
ret = db.raw_sql("""
    select permno, date, ret, prc, vol, shrout
    from crsp.dsf_v2
    where date between '1990-01-01' and '2025-12-31'
    and permno in ( ... )    -- constituent PERMNOs for the window
    """)
```

Identifiers. `permno` (permanent *security* id) is the join key everywhere; `permco` (permanent *company* id) is used to avoid treating two share classes of the same firm as an independent pair. **Universe filter.** The common-stock screen (historically share codes 10/11 and exchange codes 1–3 = NY-SE/AMEX/Nasdaq) now lives in `crsp.stksecurityinfohist` under renamed CIZ fields (`sharetype`, `securitytype`, `primaryexch`, ...); take the exact code values from the CRSP CIZ cross-reference guide. For an S&P 500-only study this screen is rarely binding (members are common stock), but it matters if the universe is later broadened.

2.3 Building the pair universe and data volume

Per window: (i) resolve constituent PERMNOs as of the window start via the spell logic; (ii) pull their daily `ret`; (iii) form all unordered pairs. With ~ 500 names a window holds $\binom{500}{2} \approx 1.25 \times 10^5$ pairs; across ~ 420 monthly windows (1990–present) that is $\sim 5 \times 10^7$ pair-windows — large but tractable. Cache the daily return panel once (a few GB), compute each window vectorized over the return matrix, and persist only the per-pair summaries ($s_{\text{odd}}, \Delta, \text{CTX}$, the four cell counts), not the raw cells. Membership churns by $\sim 20\text{--}30$ names/year, so the universe is genuinely time-varying; resolving it per window is mandatory, not a refinement.

2.4 Continuous co-membership (pair eligibility)

A pair contributes to a window only when *both* names are constituents on that window’s trading days. Because membership turns over ($\sim 20\text{--}30$ names/year, elevated in crises through forced deletions), a naive construction would (i) allow a pair’s composition to change *within* a window when a name enters or exits mid-window — contaminating the very correlations being measured — and (ii) admit short-lived pairs whose mere availability is correlated with the regime, a second-order echo of the endogeneity this analysis targets. We therefore impose **continuous co-membership**: a pair (A, B) is eligible for a window only if both A and B are constituents for the *entire* window (strict rule; a $\geq 95\%$ -of-window-trading-days threshold is the permissive variant), with the co-membership span recorded.

△ Status of this rule: a standard principle, not a branded method

There is no single citable “continuous co-membership” rule, and it would be an overstatement to present one. What *is* standard is its two ingredients, combined: (i) a *traded-throughout eligibility screen* — the formation/trading design of Gatev, Goetzmann & Rouwenhorst (2006), and the explicit requirement in later work that a security be present for the whole formation window to be eligible (reviewed in Krauss 2017); and (ii) *point-in-time* constituent reconstruction to remove survivorship and look-ahead bias. Our rule is exactly (i)+(ii) applied to *index membership* and to *pairs*, rather than to single-stock trading. Framing it this way is both defensible and accurate.

2.5 Crisis dating (kept external to the pair)

Window labels come from indicators *outside* any pair’s own returns: (i) **implied volatility** — the CBOE VIX level (high/low by a fixed or rolling threshold), available via WRDS (OptionMetrics) or directly from CBOE, and external to realized constituent returns; (ii) **NBER recession dates** — macroeconomic, fully exogenous to daily equity signs. The sample spans at least the 2000–02 dot-com unwind, the 2007–09 financial crisis, the 2011 euro-area stress, the 2015–16 drawdown, the 2020 COVID crash, and the 2022 inflation/rates repricing.

△ Why crisis dating must be external to the pair

If the crisis label were computed from the same returns being analyzed, we would reproduce at the regime level exactly the self-referential flaw this analysis exposes at the daily level. VIX comes from S&P index *options* (a forward-looking, different instrument) and NBER dates from macroeconomic aggregates — both external to any constituent pair’s own daily return path.

2.6 Cleaning and quality

(i) Drop pair-days with a missing `ret` on either leg (halts, suspensions). (ii) Define $\text{sgn}(0)$ explicitly: exact-zero daily returns occur for thinly traded names; treat $\text{sgn}(0) = 0$ and exclude those days from the ± 1 series (or fix a convention) — but be consistent and report it. (iii) Flag and down-weight ultra-thin names (many zero-volume days): their signs are stale and inflate spurious agreement. (iv) `ret` already incorporates splits and dividends, so do not re-adjust prices; note that a negative

`prc` in CRSP is a quoted bid/ask midpoint (no closing trade), so use `|prc|` if price is ever needed — though the analysis uses only `ret`, which is unaffected.

3 The cyclic-4 system on a stock pair

Outcomes and contexts. For a pair (A, B) and day t , the outcomes are the return signs $a_t = \text{sgn } R_{A,t}$, $b_t = \text{sgn } R_{B,t} \in \{-1, +1\}$. The context is the joint magnitude regime (x, y) , with $x = \mathbf{1}\{|R_{A,t}| \geq \theta_A\}$ mapped to $\{0, 1\}$ (0 = large-move, 1 = small-move) and y likewise for B . (The same construction runs with lagged-volatility regimes; the magnitude version is the sharpest case of the endogeneity and is therefore the right object for the deflation demonstration.)

Contents and connections. In CbD language the system has four *contents* (conceptual properties): q_1 = “sign of A when A is in regime 0,” q_2 = “sign of A in regime 1,” q_3 = “sign of B in regime 0,” q_4 = “sign of B in regime 1.” Each context c_{xy} jointly measures one A -content and one B -content; each content appears in exactly two contexts. This is a cyclic system of rank $n = 4$: the same algebraic shape as the CHSH experiment.

★ Bunches and connections, on a real pair

Take Deere (DE) and ADM — two S&P 500 names. Each trading day falls into one of four boxes, by whether each stock had a big or small move that day:

	ADM big-move	ADM small-move
DE big-move	$\langle ab \rangle_{00}$	$\langle ab \rangle_{01}$
DE small-move	$\langle ab \rangle_{10}$	$\langle ab \rangle_{11}$

In each box we record one number: over the window, how often do DE and ADM move the *same* direction on days that fall in that box. That single box — two stocks measured together under one condition — is a *bunch*. Now notice that “DE’s direction on its big-move days” appears in *two* boxes (top row: ADM big, and ADM small). The comparison of DE’s average direction between those two boxes is a *connection*; the four such comparisons (two for DE, two for ADM) are exactly what Δ sums. Contextuality asks: can one consistent story of how DE and ADM behave fill all four boxes at once — allowing each stock’s average direction to drift a little between boxes (that drift is Δ)? If yes, a single classical description covers everything; if the four boxes cannot be reconciled even after allowing the drift, the system is contextual.

Empirical quantities. Within each context c_{xy} , over a rolling window $\mathcal{W}$ of width w days:

$$\widehat{\langle ab \rangle}_{xy} = \frac{\sum_{t \in \mathcal{W}} a_t b_t \mathbf{1}\{(x_t, y_t) = (x, y)\}}{N_{xy}}, \quad \widehat{\langle a \rangle}_{xy} = \frac{\sum_{t \in \mathcal{W}} a_t \mathbf{1}\{(x_t, y_t) = (x, y)\}}{N_{xy}},$$

with N_{xy} the cell count, and $\widehat{\langle b \rangle}_{xy}$ analogously. Windows with $\min_{xy} N_{xy} < N_{\min}$ (we use $N_{\min} = 10$) are excluded: small cells mechanically inflate the statistics.

4 The CbD mathematics

4.1 The four CHSH combinations and s_{odd}

$$\begin{aligned} S_1 &= \langle ab \rangle_{00} + \langle ab \rangle_{01} + \langle ab \rangle_{10} - \langle ab \rangle_{11}, & S_2 &= \langle ab \rangle_{00} + \langle ab \rangle_{01} - \langle ab \rangle_{10} + \langle ab \rangle_{11}, \\ S_3 &= \langle ab \rangle_{00} - \langle ab \rangle_{01} + \langle ab \rangle_{10} + \langle ab \rangle_{11}, & S_4 &= -\langle ab \rangle_{00} + \langle ab \rangle_{01} + \langle ab \rangle_{10} + \langle ab \rangle_{11}, \\ s_{\text{odd}} &= \max_{i \in \{1, \dots, 4\}} |S_i|. \end{aligned}$$

Equivalently, $s_{\text{odd}} = \max_{xy} \sum_{xy} \pm \langle ab \rangle_{xy}$ over sign patterns with an odd number of minus signs — hence the name.

4.2 The inconsistency (direct-influence) term Δ

Each content appears in two contexts; its two versions may have different means. The total inconsistency is

$$\Delta = |\langle a \rangle_{00} - \langle a \rangle_{01}| + |\langle a \rangle_{10} - \langle a \rangle_{11}| + |\langle b \rangle_{00} - \langle b \rangle_{10}| + |\langle b \rangle_{01} - \langle b \rangle_{11}|.$$

★ What Δ measures

Δ asks: does merely *being in a different condition* shift a stock's average up/down tendency? For example, does stock A lean more negative on its large-move days than its small-move days? (It does — big down-days are bigger than big up-days.) This is ordinary “the conditions differ” behavior — real, but shallow. Cbd's point is that such shifts must be measured and budgeted for *before* any claim of deep structure.

4.3 The Kujala–Dzhafarov–Larsson criterion

For a cyclic system of rank n with dichotomous outcomes, the system is noncontextual iff

$$s_{\text{odd}} \leq (n-2) + \Delta \stackrel{n=4}{=} 2 + \Delta, \quad \text{i.e.} \quad \boxed{\text{CTX} := s_{\text{odd}} - \Delta - 2 \leq 0.}$$

$\text{CTX} > 0$ is the Cbd definition of (true) contextuality. With consistent connectedness ($\Delta = 0$) this is exactly CHSH with its classical bound 2; quantum systems reach $s_{\text{odd}} = 2\sqrt{2}$; the algebraic maximum is 4.

Σ Why the classical bound is 2 at $\Delta = 0$

Under a noncontextual (classical, common-description) model, all four contents have jointly distributed representatives $A_0, A_1, B_0, B_1 \in \{-1, +1\}$ and $\langle ab \rangle_{xy} = \langle A_x B_y \rangle$. For any fixed realization, grouping the S_1 combination as $A_0(B_0 + B_1) + A_1(B_0 - B_1)$: one bracket is ± 2 , the other 0, so each realization contributes ± 2 , and averaging gives $|S_1| \leq 2$; identically for S_2, S_3, S_4 (the single minus sign only relocates which bracket is the difference). The KDL result extends this to inconsistently connected systems: the achievable maximum of s_{odd} under the best classical stitching rises by exactly the marginal slack, i.e. by Δ .

4.4 A number you can check by hand

Σ Watch the deflation happen

Suppose one window yields these four co-movement numbers and eight conditional averages (columns are the four regimes 00, 01, 10, 11):

	00	01	10	11
$\langle ab \rangle$. (co-movement)	+0.6	+0.5	+0.5	−0.7
$\langle a \rangle$ (DE avg. direction)	−0.20	−0.05	+0.05	+0.10
$\langle b \rangle$ (ADM avg. direction)	−0.15	0.00	+0.05	+0.10

The largest combination is $S_1 = 0.6 + 0.5 + 0.5 - (-0.7) = 2.3$, so $s_{\text{odd}} = 2.3 > 2$: a naïve “violation.” But the big-move cells lean negative (the leverage effect), so each stock's average direction drifts across its two boxes:

$$\Delta = \underbrace{|-0.20 - (-0.05)|}_{0.15} + \underbrace{|0.05 - 0.10|}_{0.05} + \underbrace{|-0.15 - 0.05|}_{0.20} + \underbrace{|0.00 - 0.10|}_{0.10} = 0.50.$$

The corrected score is $\text{CTX} = s_{\text{odd}} - \Delta - 2 = 2.3 - 0.50 - 2 = -0.20 \leq 0$: *noncontextual*. The apparent violation was entirely the conditions shifting the averages — ordinary direct influence — now subtracted out. On the post-selected market data of §5 this deflation is the generic outcome.

5 The post-selection no-go for market data

The system above is structurally well-formed — it *is* a cyclic-4 system. The obstruction to interpreting $\text{CTX} > 0$ as non-classicality is that the contexts are defined by conditioning on $(|R_A|, |R_B|)$, which are functions of the very returns whose signs are the outcomes. The context is therefore *post-selected on an outcome-correlated variable* — the detection-loophole / measurement-dependence configuration. The classical bound presumes contexts assigned independently of the system (“fair sampling”); that presumption fails here by construction.

Σ A classical joint that reaches $\text{CTX} > 0$, indeed $s_{\text{odd}} = 4$

Construct a distribution for (R_A, R_B) as follows. Let the magnitude pair $(|R_A|, |R_B|)$ place positive probability on all four regime cells. Conditional on the cell, draw the sign pair (a, b) with marginals $\langle a \rangle_{xy} = \langle b \rangle_{xy} = 0$ in every cell (so $\Delta = 0$ identically) and correlations

$$\langle ab \rangle_{00} = \langle ab \rangle_{01} = \langle ab \rangle_{10} = +c, \quad \langle ab \rangle_{11} = -c, \quad c \in (0, 1].$$

Unbiased ± 1 pairs with any prescribed correlation exist (mass $\frac{1+c}{4}$ on equal signs, $\frac{1-c}{4}$ on opposite, etc.), so this is a valid, purely classical joint distribution over (R_A, R_B) . Then

$$S_1 = 3c - (-c) = 4c, \quad \Delta = 0, \quad \text{CTX} = 4c - 2 > 0 \text{ for } c > \frac{1}{2},$$

reaching the algebraic maximum $s_{\text{odd}} = 4$ at $c = 1$ — beyond the Tsirelson bound $2\sqrt{2}$, which no genuine quantum system can exceed. A fully classical process did this. Hence $\text{CTX} > 0$ on regime-sorted market data certifies nothing.

★ Why Δ cannot rescue the test here

Δ is a *first-moment* correction: it budgets for the conditions shifting each stock’s *average* tendency. But sorting days by move size distorts the *co-movement pattern* between the stocks at fixed averages — a second-moment effect that Δ never sees. In the construction above the averages are perfectly clean ($\Delta = 0$) while the sorting drives the score to its arithmetic ceiling. The leak is in the correlations, not the marginals, so the marginal correction cannot plug it.

△ The two preconditions, and where they fail

A CbD verdict reads as non-classicality only if (A) contexts are *exogenous* — assigned independently of the measured system (measurement independence); and (B) non-co-measurability arises from genuine *disturbance*, not from *selection*. Observational equity data fails both: (A) the regimes are computed from the returns; (B) a day sits in exactly one magnitude cell by sorting, not because measuring one content physically precludes another. Designed decision experiments satisfy both, which is why true contextuality has been demonstrated there (Cervantes–Dzhafarov 2018; Basieva et al. 2019) and cannot be certified here.

6 The analysis pipeline

Step 1 — Estimation. For every constituent pair and rolling window ($w = 60$ trading days, stepped monthly; θ at the per-stock rolling median of $|R|$, with fixed-quantile robustness sweeps): compute the four $\widehat{\langle ab \rangle}_{xy}$, the eight conditional marginals, s_{odd} , Δ , and $\text{CTX} = s_{\text{odd}} - \Delta - 2$. Record $(N_{00}, N_{01}, N_{10}, N_{11})$ and enforce $N_{\min}$.

Step 2 — Crisis/calm split. Label each window crisis or calm by the external indicator (§2). Report, separately by regime: the cross-sectional distribution of s_{odd} , Δ , CTX ; the proportion of pairs with $s_{\text{odd}} > 2$ (the naïve “violation rate”); and the proportion with $\text{CTX} > 0$ (the CbD-corrected rate).

Step 3 — The deflation demonstration (the central result). Fit a classical generator to

each window: a two-state regime-switching model for magnitudes with a sign-correlation that depends on the magnitude state (capturing crisis-coupling), plus an asymmetric magnitude–sign link (capturing the leverage effect); no contextual ingredient anywhere. Simulate, push through the identical pipeline, and compare the simulated and empirical ($s_{\text{odd}}, \Delta, \text{CTX}$) distributions. *Reproduction* establishes the artifact reading. Any empirical corner the classical generator cannot reach — after the controls below — is flagged, not celebrated: the presumption remains artifact until the generator class is shown inadequate.

Step 4 — Exogenous-context contrast. Recompute with contexts defined by variables *not* derived from the pair’s own returns: (i) the day’s VIX level (high/low by rolling median); (ii) scheduled macro-announcement days vs. ordinary days. Attenuation of CTX under exogenous contexts is direct evidence that the endogenous sorting was the source. (These contexts are shared, $x = y$, which empties the off-diagonal cells; the comparison therefore uses the diagonal-cell statistics.)

Step 5 — Controls. (i) Permutation placebo: shuffle regime labels within windows; the CTX distribution must collapse below 0. (ii) Small-sample null: simulate i.i.d. signs through the pipeline to quantify finite- N inflation of s_{odd} . (iii) Multiple comparisons: with $\sim 125,000$ pairs per window, control the family-wise rate (max-statistic permutation) before calling any single pair noteworthy. (iv) Sector stratification: report Steps 2–3 within GICS sectors; concentrations are candidates for the food-system follow-on, not contextuality claims.

★ How to read the final table

The headline exhibit is a two-row table — crisis vs. calm — with columns: naïve violation rate ($s_{\text{odd}} > 2$), CbD-corrected rate ($\text{CTX} > 0$), and the classical simulator’s matched rates. The expected pattern: the naïve rate is substantial and much higher in crisis; the corrected rate is far smaller; and the simulator reproduces both. That pattern, with the §5 construction beside it, *is* the deflation result: the crisis signal is real but classical — stronger regime-dependent coupling — and the quantum-looking excess is the sorting artifact.

7 Outputs and connection to the program

Deliverables: (1) the pipeline (extending the four- S codebase); (2) the deflation paper exhibit set — the table above, the CTX distributions, the classical-reproduction overlay, and the exogenous-context contrast; (3) the crisis/calm behavioral characterization (Δ , regime correlations, and their sector structure) as the aggregate System-1/System-2 contrast, stated classically. The certifiable-contextuality question proceeds in the decision/evacuation track, where contexts are experimenter-set and incompatibility is genuine disturbance; this analysis is that track’s methodological floor and its motivating contrast.

References (verified): Kujala, Dzhafarov & Larsson, *Phys. Rev. Lett.* **115**, 150401 (2015). Cervantes & Dzhafarov, *Decision* **5**, 193–204 (2018). Basieva, Cervantes, Dzhafarov & Khrennikov, *J. Exp. Psychol.: General* **148**, 1925–1937 (2019). Dzhafarov, Zhang & Kujala, *Phil. Trans. R. Soc. A* **374**, 20150099 (2016). *Pair-eligibility methodology (confirm pagination)*: Gatev, Goetzmann & Rouwenhorst, *Rev. Financ. Stud.* **19**, 797–827 (2006); Krauss, *J. Econ. Surveys* **31**, 513–545 (2017).